

CUSTOMER MILESTONE BRIEF

MS #16 - Durable Factory Recovery Control Plane

Purpose, customer value, delivered capability, evidence, and claim boundaries

Status	Complete
Release	0.16.0
Date	2026-08-15
Audience	Customers, partners, executives, architects, delivery teams, and assurance reviewers

Executive summary

MS #16 - Durable Factory Recovery Control Plane converts a specific part of the LIGHTYEAR modernization approach into a governed, reviewable capability. Makes long-running factory work recoverable across interruption, restart, and controller failure.

Why this matters to a customer

Makes long-running factory work recoverable across interruption, restart, and controller failure.

The practical result is a narrower, evidence-backed decision instead of a broad modernization claim. Commercial, architecture, delivery, security, and assurance teams can review the same capability, proof, and limits.

What the milestone delivers

- Added a transactional durable control plane with a SQLite WAL reference backend and explicit contracts suitable for PostgreSQL and immutable object-store production adapters.
- Added atomic worker leasing, opaque bearer tokens stored only as SHA-256 digests, bounded heartbeats, lease expiry recovery, retry backoff and terminal dead-letter handling.
- Added crash-safe wave barriers and idempotent completion: passed cells are never dispatched twice and successor waves remain unavailable until every predecessor has passed.
- Added exactly-once consumption of human portfolio approvals, exact work-order identity checks, content-addressed receipt indexing and a tamper-evident durable event chain.
- Added cross-platform durable queue commands, versioned lease/state/snapshot schemas, operator guidance and a strictly read-only Recovery Control Tower with no dispatch authority.
- Added concurrent lease, worker termination, stale-token, replay, retry, dead-letter, wave-barrier, artifact-index and event-tampering tests plus full verification integration.

How it differs from earlier work

MS #16 builds on MS #15 (Conflict-Aware Portfolio Factory). The earlier milestone established its own bounded capability; this milestone adds durable factory recovery control plane while preserving the earlier evidence and its limits.

Earlier evidence is not automatically promoted into this milestone. A parser, simulator, local comparison, or generated artifact is not live platform equivalence or production readiness.

What a customer can do with it

A customer can use durable factory recovery control plane as a bounded decision and delivery capability, attached to approved sources, owners, policies, target architecture, acceptance criteria, and authorized evidence.

Unresolved semantic loss, policy decisions, unsupported behavior, and live-evidence requirements remain visible.

Evidence and acceptance posture

The repository capability is complete because its committed implementation and release record are present. Customer qualification remains claim-specific and evidence-class-specific.

Review exact source and artifact identities, distinguish development from authorized live evidence, inspect policy and loss classifications, confirm passed and blocked gates, and verify that later changes have not invalidated the evidence.

Boundaries and claims not made

- This milestone establishes only the bounded capabilities and evidence listed here; later capabilities are not retroactive.
- Production readiness and platform equivalence require the explicit evidence and policy gates applicable to the customer environment.

Source of record

- CHANGELOG.md release section(s): 0.16.0
- README.md product and operating guidance
- LIGHTYEAR-ROADMAP.md governing sequence and claim classifications
- docs/milestones/catalog.json metadata and docs/milestones/manifest.json artifact integrity